

**LUNDS TEKNISKA HÖGSKOLA**  
**Inst. för telekommunikationssystem**

**Tentamen i *DATORKOMMUNIKATION***

**Datum: 2005-12-15**

**Tid: 14 – 19**

**For tasks 1-5: you must NOT use the text book. For tasks 6-8: you may use the text book**

**Solutions for mid term exam 3 and tasks 1-5 should be handed in before you may use the book!**

**When the text book is used, a red A4 sheet should be placed at your table!**

**For those of you that do not have the text book, you may borrow the text book from us.**

**However, you may only read the book at a designated place in the room!**

**The total exam is judge as 3: 50-64 cr, 4: 65-79 cr, 5: 80 cr or more.**

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**TASKS 1-5, the text book must not be used!**

**Task 1 (10 cr)**

- a) What are the concerns of the physical layer in the Internet model?
- b) What are the three sets of redundancy checks used in data communications?
- c) What is the difference between a root bridge and a designated bridge?

**Task 2 (10cr)**

- a) What is the purpose of ARQ mechanisms?
- b) How does Go-Back-N ARQ differ from Selective Repeat ARQ?
- c) What is the purpose of the timer at the sender side in systems using ARQ?
- d) Which ARQ mechanisms utilize pipelining?
- e) How is the bandwidth-delay product related to the system efficiency?

**Task 3 (10cr)**

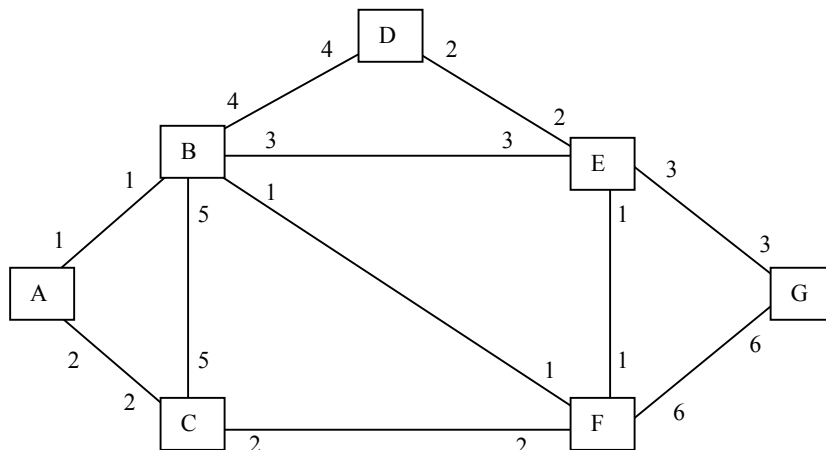
- a) The Data Link layer in LANs is often subdivided into two sublayers, the Logical Link Control sublayer (LLC) and the Medium Access Control sublayer (MAC). Describe the functions for the
  - LLC sublayer
  - MAC sublayer
- b) The CSMA/CD is a MAC sublayer protocol. Describe how it works.
- c) Compare Controlled based MAC protocols and Random Access based MAC protocols.
  - Under which network loads (high or low) operates the one better than the other?
  - If we wish to give QoS in terms of short delays, which of the two should we use?

**Task 4 (10cr)**

- a) The Internet Protocol (IP) is an unreliable and connectionless datagram protocol – a best-effort delivery service. What is here meant by:
  - unreliable
  - connectionless datagram
  - best-effort delivery service
- b) The next-generation IP (IPv6) has some advantages over IPv4. Give three examples of such advantages.
- c) TCP is a reliable transport protocol.
  - What is here meant by a reliable transport protocol?
  - Give the three most important functions of the TCP protocol.

Task 5 (10 cr)

- Describe the individual steps in the Dijkstra Algorithm!
- Use the Dijkstra Algorithm to find the shortest path tree from **node A** in the graf below. Answer with the resulting tree, including costs.



- State the resulting routing table for node A.

**TASKS 6-8, the textbook may be used!**

Task 6 (15 cr)

Consider the local area network given below. Two frames, (originally) transmitted from node A, have been captured by an Ethereal sniffer. MAC-addresses for the different nodes are known:

- node A: 00 08 74 41 AF A7
- node B: 00 08 74 41 EA 0A
- node C: 00 00 39 51 90 37
- node D: 00 D0 C0 21 28 00

Study the two frames given below and determine, if possible, the object of each node A to D. The frames are Ethernet-II traffic, where the Preamble, SFD, and CRC fields have been removed. Both frames carry IP-packets.

Hint: Object suggestions: host, router interface, telnet server, DNS server, SMTP server.

Frame 1:

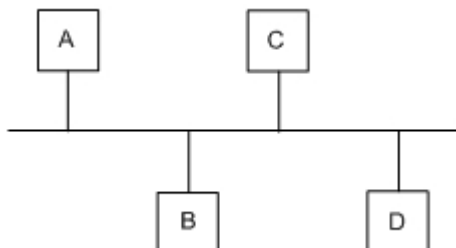
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00 00 39 51 90 37 00 08 74 41 AF A7 08 00 45 00 05 DC 48 00
00 00 20 11 D2 A9 B6 03 06 48 B6 03 08 19 04 01 00 35 ...
  
```

Frame 2:

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00 D0 C0 21 28 00 00 08 74 41 AF A7 08 00 45 00 0A 23 36 00
00 00 80 06 82 3C B6 03 06 48 B6 04 06 49 09 93 00 17 ...
  
```



### Task 7 (15 cr)

- a) Given a network with 6 subnets, numbered from 1 to 6, and 4 routers, numbered from A to D. The routers use a Distance Vector routing protocol. Each subnet is connected to max 2 routers. Subnets 1, 2, and 3 are stub nets. Below is given the routing tables of routers A and D. Draw a sketch of the complete network and mark out the subnets and the routers. Hint: Think of a subnet as one Ethernet; there are no routers inside the subnet.

| Router A routing table |      |          |
|------------------------|------|----------|
| Subnet                 | Cost | Next-hop |
| 1                      | 1    | -        |
| 2                      | 4    | D        |
| 3                      | 2    | D        |
| 4                      | 3    | D        |
| 5                      | 2    | D        |
| 6                      | 1    | -        |

| Router D routing table |      |          |
|------------------------|------|----------|
| Subnet                 | Cost | Next-hop |
| 1                      | 2    | A        |
| 2                      | 3    | C        |
| 3                      | 1    | -        |
| 4                      | 2    | C        |
| 5                      | 1    | -        |
| 6                      | 1    | -        |

- b) The networks layout is changed and router A receives a routing update from a new router E. Show router A's routing table after the update.

| Routing update from router E |      |          |
|------------------------------|------|----------|
| Subnet                       | Cost | Next-hop |
| 1                            | 1    | -        |
| 2                            | 1    | -        |
| 3                            | 2    | B        |
| 4                            | 4    | B        |
| 5                            | 3    | A        |
| 6                            | 2    | A        |
| 7                            | 2    | B        |

- c) Draw a sketch of the networks new layout.

### Task 8 (10cr)

- a) You have a 10Base-T Ethernet segment and a 100Base-TX Ethernet segment. Can these two segments be connected by a
- Repeater? Motivate your answer!
  - Bridge? Motivate your answer!
- b) A router connects a 10Base-T Ethernet with a network that utilizes Token Passing as the MAC protocol. Both networks are IP-based and the users in the two networks just use the TCP or UDP protocols as transport protocols. Describe the Physical, the Data Link and the Network layers at the two sides of the router in detail.
- c) What Internet model layers are of concern to LANs?

Good luck

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